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RESEARCH ARTICLE

Deep Learning for Dust Accumulation Analysis on Desert Solar Panels: A CNN-Transformer Approach

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ABSTRACT In the energy transition towards sustainability, photovoltaic power is increasingly valued for its eco-friendly and renewable attributes. Northern and northwestern China's deserts, abundant in solar energy and vast in land, are prime areas for photovoltaic generation. High winds and dust storms in these regions cause dust buildup on PV panels, reducing efficiency and potentially harming the equipment. In water-scarce deserts, cleaning photovoltaic panels is particularly difficult. Therefore, developing robust and precise dust accumulation detection techniques is crucial to ensure the stable operation of PV power plants in deserts. To address this challenge, this study proposes a solution based on the convolutional neural network (CNN) structure UTran-ResNet50, a model capable of accurately recognizing images with various levels of dust coverage on photovoltaic panels. By combining ResNet50 and Transformer architectures, their strengths in global and local feature extraction are effectively exploited. In addition, the study incorporates the Attention U-Net module to reduce the recognition error and introduces the Positional Attention module to enhance the feature recognition capability. It is particularly noteworthy that the experimental validation conducted with a homemade dataset, despite its limited sample size, demonstrates the high efficacy of the proposed network, achieving an accuracy of 96.41%. It exemplifies the model's robustness in dealing with small sample datasets, highlighting its superior performance in accurately detecting dust accumulation on photovoltaic panels.

INDEX TERMS Deserts, PV panels, dust accumulation, CNN, transformer.

I. INTRODUCTION

Energy and environmental issues have emerged as the two primary challenges hindering sustainable economic and social development globally. As the global population expands and industrialization accelerates, traditional fossil energy sources are increasingly unable to meet the rising energy demand, while the carbon dioxide emitted from their combustion significantly contributes to global warming [1]. The ongoing depletion of fossil energy sources, coupled with escalating energy costs, has intensified interest in clean and

renewable energy alternatives [2]. Consequently, researchers and policymakers are actively pursuing sustainable energy strategies. Energy research represents not only a technological challenge but also a vital pathway to achieving sustainable development, with improvements in energy efficiency serving as an effective means to reduce both energy consumption and environmental impacts [3].

Solar energy, recognized as a new green energy source, is favored for its renewable and clean characteristics [4]. Recent advancements have made the direct conversion of solar energy into electricity a rational and efficient method of energy utilization. The efficiency of photovoltaic (PV) modules has significantly increased, while their

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manufacturing costs have decreased. The desert regions of northern and northwestern China, characterized by high radiation, expansive land, and low land costs, have emerged as key potential areas for the deployment and efficiency optimization of PV technology, as well as important development zones for concentrated solar power generation [5], [6], [7]. The surface morphology in these regions is predominantly sandy, making it susceptible to floating dust and dust accumulation when exposed to high winds and dust storms, resulting in the deposition of sand particles on the surfaces of photovoltaic panels. The accumulation of dust not only reduces the light transmission of PV modules, thereby affecting power generation efficiency [8], [9] but can also trigger the hot spot effect, which damages the PV panels and shortens their service life. Consequently, the cleaning and maintenance of photovoltaic panels have become critical issues, with the main cleaning methods currently available including natural, manual, and mechanical approaches [10]. However, water scarcity and cost constraints in desert areas limit the feasibility of manual and mechanical cleaning, while natural cleaning methods are contingent on weather conditions. Considering these challenges, techniques for accurately monitoring dust accumulation on photovoltaic panels have become particularly crucial. Developing an efficient method to detect uneven dust accumulation on photovoltaic panels is vital, as it facilitates the implementation of timely cleanup strategies, ensuring the stable and efficient operation of photovoltaic systems in desert environments.

With the emergence of deep learning (DL) techniques, convolutional neural networks (CNNs) have performed remarkably in processing data characterized by grid-like structures, such as images and videos. This advancement facilitates the automatic and effective extraction of critical information through the recognition and comprehension of local features and spatial hierarchies inherent in visual data. In the field of image recognition and classification, the system can learn spatial hierarchical features from input images in an unsupervised and autonomous manner and continuously refine its recognition ability through adaptive learning [11], [12], [13], [14], [15], [16]. In this work, we exploit these advantages of CNNs to propose a UTran-ResNet50 network architecture, which combines the benefits of global and local feature detection to enhance the network's ability to understand complex patterns in data. The effectiveness of UTran-ResNet50 is demonstrated in the image classification task of detecting dust accumulation on the surface of the panels, and the experimental evaluation results prove its excellent performance in accurate detection and classification.

In summary, the principal contributions of this paper are as follows:

- Centering and segmenting solar panels in images mitigate weight shift from uneven placement, boosting validation efficiency with minimal samples, while the convolution, decoding, and centering processes minimize computational errors for accurate network data.

- The network architecture of ResNet50 is retained while the feature extraction capability is enhanced. By reducing the spatial size of the features, increasing the number of channels, and introducing an attention mechanism, the network's ability to extract meaningful information from the input data has been improved.
- The introduction of the Positional Attention mechanism helps to maintain the integrity of the spatial information more accurately and, in combination with the Transformer's multi-attention mechanism, enables an effective focus on establishing dependencies over long distances. The model has been refined to detect even the subtlest changes in dust accumulation on solar panels with enhanced precision.

II. RELATED WORK

A. PV EFFICIENCY AFFECTED BY DUST

Numerous studies have demonstrated that the composition, nature, and deposition rate of dust exhibit global disparities. These differences directly influence the extent to which photovoltaic (PV) efficiency is compromised, underscoring the necessity for region-specific models and cleaning strategies to maximize energy output. In addition, the effect of dust on PV efficiency is influenced by geographical location, climatic conditions, and environmental factors [17]. In a study conducted in Malaysia [18], the peak power output of photovoltaic (PV) systems was observed to decrease by 18% due to dust accumulation. An experiment carried out in Indonesia [19] revealed that dust on PV panels led to a 10.8% reduction in power output, corresponding to an average relative humidity of approximately 52.24%. Kazem et al. [20] found that, under desert conditions, the efficiency of photovoltaic panels can decrease from 16% to 8% within 45 days due to dust accumulation. Klugmann-Radziemska [21] showed a clear linear correlation between dust accumulation and efficiency loss in crystalline silicon PV modules in northern Poland. Just 0.862 μm of dust reduced short-circuit current from 4.80 A to 4.16 A, highlighting the need for dust management in PV systems. In addition, dust samples were collected from six different sites in northern Oman for in-depth analysis, and soiling losses from PV modules left uncleaned for three months resulted in a 35-40% reduction in power output [22]. Said et al [23] studied the current-voltage (I-V) curves of two identical monocrystalline photovoltaic modules, which were adjustable to eight varying inclinations to simulate environmental exposure. One module was regularly cleaned, and the other was naturally dusted. It was found that 5 months of natural dusting resulted in a 6% loss of module power. Enaganti et al [24] investigated dust accumulation on PV modules placed at different angles and found that geographic dust characteristics have a direct impact on module dust accumulation and cell efficiency. Therefore, dust detection is essential to ensure panel cleanliness.

B. PHOTOVOLTAIC DUST ACCUMULATION DETECTION METHODS

To quantify dust accumulation on solar panels, researchers have employed a variety of techniques, including direct measurement, thermal imaging, image processing, Internet of Things (IoT) technology, and artificial intelligence algorithms. Table 1 outlines detection methods and their characteristics. In laboratory settings, a commonly utilized method involves collecting dust from the surface of photovoltaic (PV) panels using distilled water. This is followed by filtration to separate the dust, which is then dried and weighed to determine its mass. The concentration of dust is indirectly obtained by comparing the weight difference between clean and polluted PV panels [9]. Reference [25] proposed an alternative approach involving wiping the PV plate with a dry sponge and calculating the dust concentration based on the weight difference before and after cleaning. However, this method may yield inaccurate measurements due to the potential loss of dust during movement. While offering some solutions, these methods often suffer from inefficiencies or lack of precision. The remote detection capabilities afforded by thermal imaging enable effective operation even in challenging access situations or harsh environmental conditions [26]. Hotspots generated by dust deposition can be detected using thermal imagers; their application enhances both the speed and accuracy of inspections while minimizing additional damage to PV modules during the process. Jeffery et al. [27] utilized infrared thermography (IRT) to monitor solar module status and intelligently classify faults through fuzzy logic assessment of thermal conditions. This approach allows for automatic diagnosis by analyzing thermal images from faulty versus healthy panels. Cubukcu and Akanalci [28] identified and localized defects in photovoltaic modules through thermographic analysis across 19 different photovoltaic power systems in Turkey—demonstrating significantly greater effectiveness and efficiency compared to other commonly used inspection methods.

Conversely, the Internet of Things (IoT) and sensor-based technologies are employed to detect and periodically clean photovoltaic panels. Mohammed et al. [29] investigated the photovoltaic (PV) power generation efficacy within the Sahara Desert and engineered a sensor-assisted measurement apparatus tailored for arid settings, wherein particulate matter is the predominant influence on the performance of PV systems in regions characterized by intense irradiance and copious sand and dust. This apparatus is designed to initiate automated cleaning procedures when the power output falls below 50%. Smestad and associates [30] formulated a model for quantifying PV performance degradation due to dust accumulation via optical assessment, uncovering a direct proportional correlation between the extent of dust coverage and the transmittance ratio. The application of transmittance ratios enhanced the precision of dust concentration estimation. In separate research, Zainuddin et al. [31] proposed a real-time surveillance technique for solar cells leveraging

the Internet of Things (IoT) and employed a smartphone application to track the energy produced and to facilitate the cleaning of the PV surface as required. Nonetheless, these methodologies are confined to measurement or application in specific regions and are influenced by the conditions of the installation setting.

In recent years, as technology continues to advance, many scholars have used artificial intelligence methods to measure and detect dust accumulation. Huang et al. [32] created the ABC-SSELM algorithm for diagnosing PV panel faults with limited labeled data, enhancing accuracy and reliability, with future work focusing on its application for dust monitoring to improve cleaning efficiency. Saquib et al [33] constructed an artificial neural network (ANN) to predict the output voltage and current using information from pictures of dust on solar panels. The network contains a hidden layer of 9 neurons with irradiance and amount of dust as input parameters. For the same purpose, Mehta et al [34] used a convolutional neural network (CNN) combined with bi-directional input-aware fusion (BiDIAF) to classify dust-affected regions and predict functional loss, achieving about 3% prediction accuracy and 4% localization accuracy, and boosting localization studies by 24% in weakly supervised learning. Fan et al. [35] introduced a deep residual neural network (DRNN) for identifying non-uniform dust accumulation on photovoltaic (PV) panels. They optimized the network's architecture by linking residual elements via skip connections, thereby reducing the order of the weight matrix. This modification enhanced the network's adaptability and the precision of feature learning. Another study has proposed an improved hybrid solar cell defect detection method that utilizes Generative Adversarial Networks (GANs) and a weighted classification method, aiming to enhance the accuracy and efficiency of solar cell defect identification [36]. Onim et al. [37] presented a convolutional neural network (CNN) model of SolNet dedicated to the detection of solar panel dust accumulation. Its efficient and accurate detection capability was verified by training and testing a dataset of clean and dust-accumulated solar panel images. Alatwi et al. [38] used deep learning techniques to detect dust accumulation on solar panels, proposing a low-cost, sustainable solution to increase solar power generation by predicting voltage using image processing. Shao et al. [39] proposed a new method for detecting dust on the surface of PV panels based on PyTorch and improved the detection efficiency by optimizing the Adam algorithm and introducing the warm-up strategy and cosine annealing strategy, which are of great importance for the maintenance and performance optimization of PV systems.

III. METHODS

The world has seen significant changes with neural networks, where models can learn and make predictions. These models, trained on specific datasets, often struggle with category imbalance and identification of key dust features. Traditional CNNs are known for their strong local feature extraction

TABLE 1. Detection methods and characteristics.

Types of Testing Methods	Characteristics	Reference
Lab Direct Measurement	Weight Difference for Dust Quantification	[9] ,[25]
Thermal Imaging	Infrared Detection of Dust Hotspots	[26] - [28]
IoT Sensors	Automated Cleaning via Threshold Triggers	[29],[31]
Optical Model	Transmittance-Dust Coverage Correlation	[30]
Deep Learning	CNNs, GANs, etc. for precision detection	[32]- [39]

capabilities, but with the continued development of deep learning, global information extraction capabilities have been significantly improved. In contrast, the Transformer model incorporates positional encoding in the input token, which is more effective in dealing with details and local features. Unlike previous studies, this paper uses a homemade dataset to classify it into five different dust accumulation classes for analyzing solar panel dust, processes the images with a balanced layout for effective model validation even when there are fewer samples, and proposes the architecture of UTran-ResNet50, which is specifically used for grading dust accumulated on the surface of photovoltaic panels in desert environments. The current algorithm is applied, and the performance is compared with other algorithms with an accuracy of 96.41%. The algorithm is essential for grading dust accumulation for maintenance strategies and safeguarding solar panel performance.

A. OVERALL NETWORK STRUCTURE

In this study, we introduce UTran-ResNet50, a novel deep-learning architecture that integrates ResNet50 with a Transformer in a sequential manner to create an enhanced network design. Both local feature details and global contextual information are included. This fusion strategy allows the system to understand the image content more comprehensively and improves the classification accuracy.

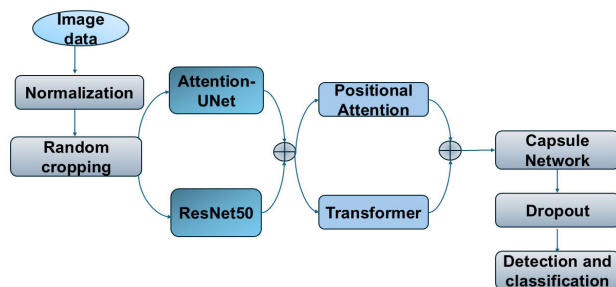


FIGURE 1. The overall architecture of the proposed network.

As shown in Fig. 1, the network consists of two main core parts; the first one is composed of the Attention U-Net

module and the CNN coding branch. The CNN branch uses ResNet50 as the backbone network to extract raw features. Attention U-Net is mainly used to extract local features and perform pixel-level prediction, while ResNet50 focuses on capturing high-level semantic information. These two modules are passed to the next layer by either concatenation or weighted summation and then passed to the next layer. The second component of the model employs a Positional Attention mechanism and the parallel architecture of the Transformer, addressing the issue of vanishing gradients associated with deep networks. This approach effectively preserves spatial information, while the Transformer module is adept at capturing long-range dependencies. On the one hand, CNN can capture the local features of the image through convolutional layers and reduce the spatial dimensions of the features through pooling layers to achieve the extraction of global features of the image. On the other hand, the Transformer structure captures scenes with long-range dependencies and more finely captures local features of accumulated dust in the image. This integrated approach leverages the strengths of both components, thereby enhancing the model's ability to achieve more refined characterizations.

B. ATTENTION U-NET

Attention-U-Net improves the performance of the original U-Net in several ways by introducing an attention mechanism, especially in tasks that require accurate segmentation and good background suppression [40]. Attention gates are incorporated into the skip connections to dynamically adjust the feature weights from the encoder based on information from the decoder side, thereby determining which encoder-side features should be enhanced and allowing the network to focus more on regions that are beneficial for the current segmentation task. We employ this module in the processing of image datasets by first applying a panel-like process to the input image through a convolutional operation, a process that can be viewed as applying a series of filters to the image to extract local features and texture information from the image. After the convolution operation, the image features are encoded into higher-dimensional feature representations. Subsequently, these feature representations are remapped back to the original image space through a decoding process to recover the structural information of the image while minimizing the information loss. To further improve the accuracy of image processing, the module adjusts the decoded image with center offset, ensuring that the key features are in the geometric center of the processing flow by performing a panning operation on the image, which helps to improve the model's ability to identify and segment the dust accumulated on the surface of PV modules, thus reducing the computational error and optimizing the feature extraction and classification performance of the subsequent network layer. Fig. 2 illustrates the architecture of the Attention U-Net block.

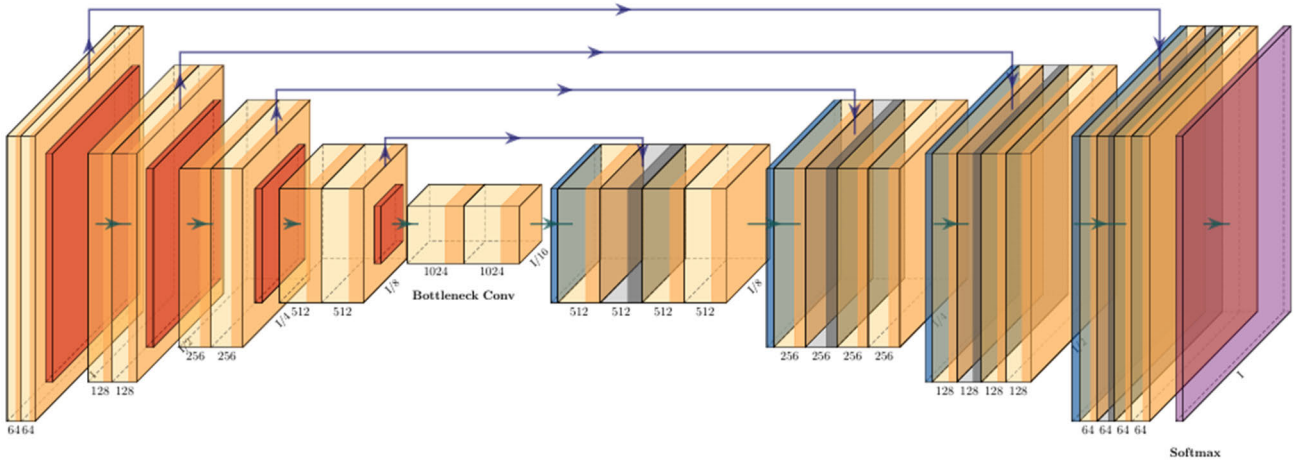


FIGURE 2. An illustration of the Attention U-Net architecture.

C. CNN ARCHITECTURE

ResNet50 was chosen as the backbone network of the CNN coding branching system, and the pre-trained ResNet50 network is utilized for feature extraction. The deep residual architecture effectively mitigates the issues of gradient vanishing and gradient explosion, thereby facilitating the network's ability to learn feature mappings more effectively. This not only reduces the training time but also lowers the overall training complexity. Fig. 3 illustrates the architecture of ResNet50. Its network creates a fully connected layer and includes multiple Residual Blocks. Each Res Block consists of two convolutional layers, short-circuiting of features is achieved through jump connections, which helps to maintain the gradient flow during training, allowing the network to learn deeper feature representations and have the same number of channels. This design enhances the feature learning capability while maintaining feature dimensionality. Residual connectivity facilitates inter-block communication with the reconstruction module, enabling the aggregation of multi-level features.

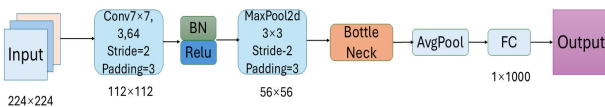


FIGURE 3. Overview of ResNet50.

The convolutional layers within the residual blocks are utilized to improve the feature extraction efficiency. To reduce the initialization sensitivity of the convolutional network and speed up the operational efficiency of the network, a batch normalization layer is added after the convolutional layer, which is calculated as shown in equation (1):

$$x_b^* = \frac{x_i^j - \mu_b}{\sqrt{\sigma_b^2 + \varepsilon}} \quad (1)$$

where x_i^j is the i -th value in the j -th batch, x_b^* is the batch normalized value, μ_b is the mean value of the current batch, σ_b indicates the standard deviation of the current batch, ε is a very small positive number set to prevent the denominator from being zero.

We use preprocessed images as inputs, which are passed at the same resolution through multiple residual blocks in the ResNet50 network that can generate feature representations of varying depths. The network is initialized with pre-trained weights to expedite the training process and enhance its generalization ability, while its design is optimized for the enhanced relocation and simplification of features, enabling the extraction of multi-scale features. At the tail of the network, the feature map is compressed into a one-dimensional feature vector via a global average pooling layer, followed by a fully connected layer, which maps the feature vector to the category space and outputs a probability distribution corresponding to the number of categories.

In addition, the Dropout layer is used for regularization to prevent overfitting. To optimize the model performance, an Adam optimizer and cross-entropy loss function were used for training. These features were then passed to the Attention module for further processing and analysis.

ResNe50 incorporates the Bottleneck building block, as depicted in Fig. 4, which is designed with a reduced parameter count and enhanced computational efficiency, thereby preserving the network's performance.

D. POSITIONAL ATTENTION

To effectively capture sequence dependencies, we introduce the Positional Attention module, which incorporates broader contextual information into local features for enhanced representation [41]. Next, we elaborate on the process of adaptively aggregating spatial contexts. As shown in Fig. 5, given a localized feature map A, three feature maps {B, C, D} are first obtained by three convolutional layers (BN and ReLU) respectively, shaped to (C*H*W) and then reshaped

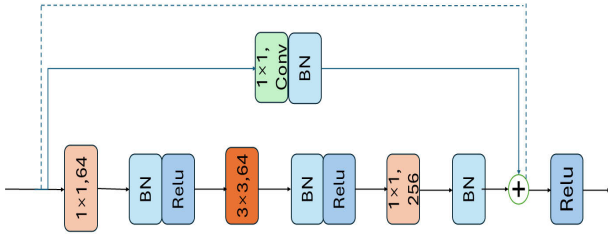


FIGURE 4. Architecture of the bottleneck.

to $C \times N$. Matrix multiplication is performed between the transpose of matrices C and B . In multiplying the output by SoftMax with the transpose of the matrix D , reshape the result to $(C \times H \times W)$ and then multiply it by the scale factor α , and then reshape it to the original shape $(C \times H \times W)$, and finally add it to A to get the final output, which is obtained by SoftMax with each element as equation (2).

$$S_{ji} = \frac{\exp(B_i \cdot C_j)}{\sum_{i=1}^N \exp(B_i \cdot C_j)} \leftarrow \quad (2)$$

where s_{ji} is the effect of location i on location j , feature map A into the convolutional layer to generate a new feature mapping D , which is reshaped. D and the reshaped transpose and shape the result to $\mathbb{R}^{C \times H \times W}$, scaled by α and perform an element-by-element summation operation on the feature A to obtain the final output where each element E is:

$$E_j = \alpha \sum_{i=1}^N (S_{ji} D_i) + A_j \quad (3)$$

where α is the scale factor, which is initialized to 0 and gradually learns to assign more weights. From equation (3), it can be inferred that the resultant feature at each location is a weighted sum of the features at all locations and the original feature. Thus, it has a global contextual view and selectively aggregates contexts based on spatial attention maps. Similar semantic features gain from each other, thus improving intra-class compactness and semantic consistency.

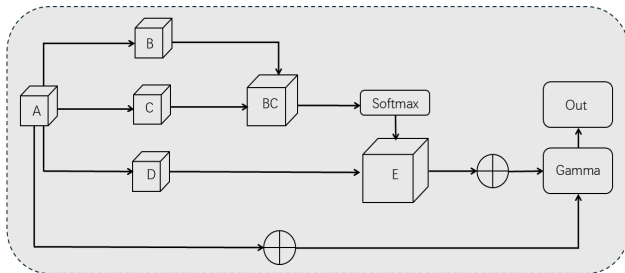


FIGURE 5. An illustration of positional attention module.

E. TRANSFORMER NETWORK STRUCTURE

In this system, the Transformer model is used to extract the local features of the image [42]. It mainly consists of a self-attention mechanism, Multi-Head-Attention (MHA), and a

Feed-forward Neural Network (FFN). The self-attention mechanism processes the image, which will be the input sequence after a linear change into three matrices, i.e., the query (Query, Q), the key (Key, K), and the value (Value, V), which is computed as shown in equations (4) and (5), Fig. 6 Schematic diagram of the self-attention mechanism. The multi-head self-attention mechanism obtains a more comprehensive understanding of the original data by dividing the input into multiple subspaces, calculating attention distributions for each subspace separately, and then merging these distributions, thereby performing separate self-attention computations according to formulated attention heads, capturing diverse relationship patterns and feature representations, and mapping them to the output space after output splicing and linear transformation. The mathematical expression of multi-head self-attention is shown in equations (6) and (7).

$$Q = XW^Q, K = XW^K, V = XW^V \quad (4)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (5)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^0 \quad (6)$$

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \quad (7)$$

where X is the image sequence, W is the weights, Q , K , V are the query matrix, key matrix and value matrix obtained after the input vectors go through the linear projection layer, d_k denotes the dimension of the key matrix, and h denotes the number of attention heads.

The self-attention score matrix is first obtained by performing a dot product operation with and dividing by the scaling factor, to obtain the self-attention score matrix. Next, each single-head self-attention is subjected to attention weight calculation separately to obtain the attention output which represents the correlation between the information of the input sequence. Finally, the results of all single-head attention calculations are spliced and linearly transformed to obtain the final output.

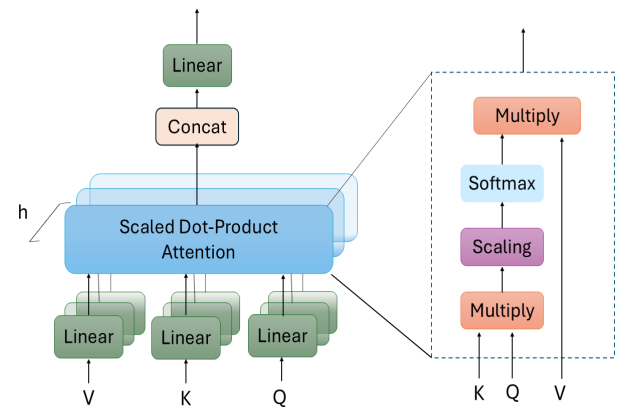


FIGURE 6. Self-attention mechanism.

The feed-forward neural network layer consists of two linear transformation layers connected with a ReLU activation function in the middle, which is mainly used for the multi-head self-attention mechanism and mapping the output connected with the residuals into the high-dimensional space, and mapping more detailed information in the high-dimensional space into the low-dimensional space through the linear layer to mine deeper features. The specific formula is shown in equation (8):

$$FFN(x) = ReLU(0, xW_1 + b_1)W_2 + b_2 \quad (8)$$

where W_1 and W_2 are the weight matrices of the two linear layers, b_1 and b_2 are the bias vectors of these two layers.

IV. RESULTS AND DISCUSSION

A. DUST ACCUMULATION CLASSIFICATION EXPERIMENT AND METHOD

To simulate the real situation of dust accumulation on the surface of photovoltaic panels in a wind-sand environment and to reduce the influence of soil particle size on the experimental results, this study sampled the surface dust of photovoltaic panels in a photovoltaic power station in Xinhua Forest Farm, Linhe District, Inner Mongolia Autonomous Region, China. The power station is in a sandy soil area, and the ground is mainly composed of fine sand and gravel, whose loose nature makes the soil easy to erode by wind erosion. Sampling in this environment helps to improve the accuracy and reliability of the experiment. The field map is shown on Fig. 7.



FIGURE 7. Field sampling locations.

Before the experiment, the dust accumulation samples were placed in an industrial drying oven and dried for 24 hours. Subsequently, the samples underwent particle size screening, utilizing a 60-mesh sieve to effectuate a refined filtration of the samples. A closed experimental environment was set up to simulate the actual dust accumulation process. In this environment, a blower with a speed of 1,000 r/min was utilized to generate a uniform airflow, which was operated continuously until the dust was completely lifted in the airtight container. The PV panels were left to stand for 5 hours in a dry environment to allow the dust particles to naturally deposit and uniformly adhere to the glass surface of

the PV panels, thus obtaining samples with different levels of dust accumulation. During the dust accumulation phase, a smartphone camera was used to capture a series of images at various times of the day, thereby encompassing a wide range of lighting conditions. The glass panels were oriented at angles ranging from 15° to 40° to simulate the diverse tilting scenarios that might occur in real applications. This method facilitated the creation of a comprehensive dataset of dust accumulation imagery under varying tilt angles and illumination conditions. After the image capture phase, the transmittance of the glass panels was quantitatively measured using a THC-100 color haze meter. The transmittance data obtained were systematically analyzed, leading to the classification of samples into five distinct transmittance categories: 'Very Clean,' 'Slightly Clean,' 'Medium Clean,' 'Slightly Unclean,' and 'Very Unclean.' These categories were defined based on transmittance thresholds of >85%, 75–85%, 65–75%, 55–65%, and <55%, respectively. Detailed records of the experimental parameters were meticulously maintained for each transmittance category. Thereafter, the collected data and images were transferred to a computer system for further image processing analysis. The key steps of the experimental procedure are depicted in Fig. 8.

B. DATASET DESCRIPTION

Based on experimental procedures, a dataset consisting of images of glass panels with varying levels of dust deposition and their corresponding transmittance data was systematically constructed. The dataset encompasses five categories, with a total of 836 images. Table 2 lists the number of images in each category. All images were uniformly resized to a resolution of 1024 × 1024 pixels, and grayscale normalization was applied to ensure consistency and comparability across the dataset. To ensure the statistical validity of the data distribution, hierarchical sampling was utilized to divide the dataset into a training set (669 images) and a validation set (167 images) at an 8:2 ratio. This method ensured that the proportion of each category in the subset mirrored that of the original dataset, thereby minimizing category bias. Additionally, to enhance the model's generalization capability, data augmentation techniques were applied to a subset of the training images. These techniques included random rotation, Gaussian blur, brightness and contrast adjustments, local occlusion, and noise superposition simulating diverse climatic conditions. Such methods were implemented to strengthen the model's robustness in identifying dust accumulation levels under varied environmental scenarios.

C. EXPERIMENTS SETTINGS

The experimental implementation of the proposed model utilizes the TensorFlow 2.18.0 deep learning training framework along with the Keras open-source library and the Python programming language. The Adam optimizer is used for training with the default learning rate and the categorical cross-entropy loss function. The system used for the

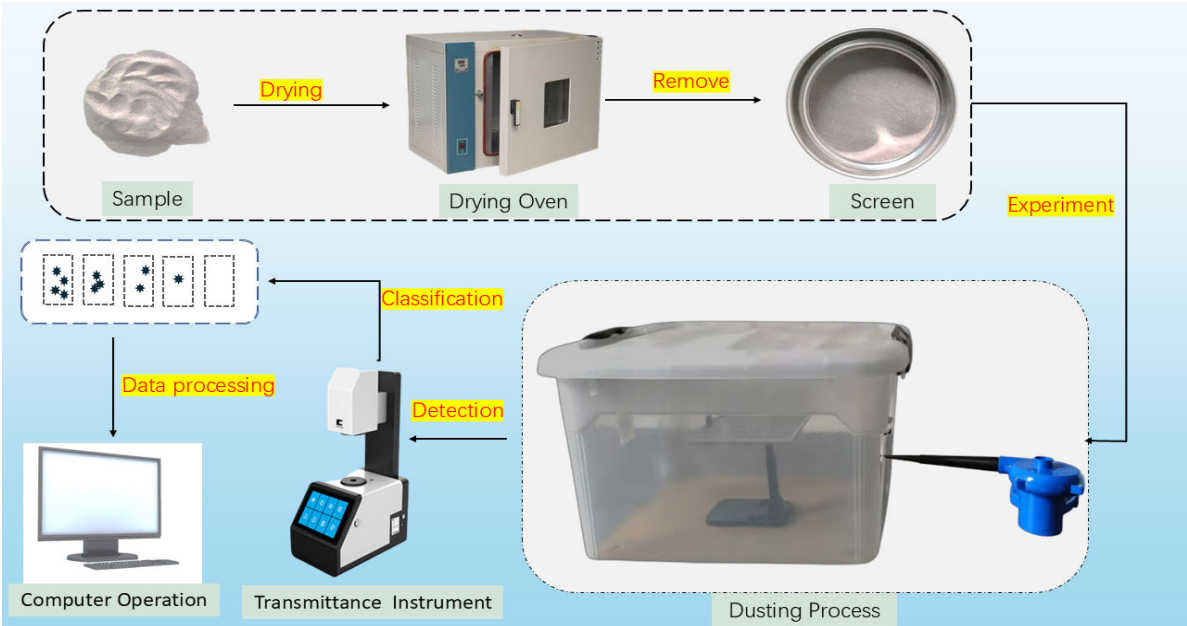
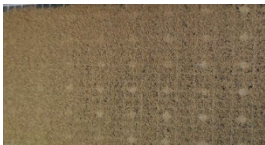
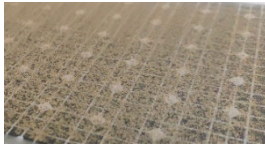
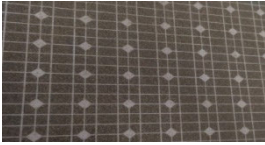
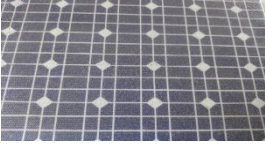
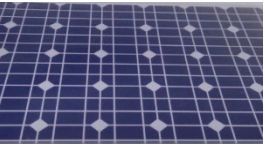


FIGURE 8. Experimental procedure.

TABLE 2. Sample images and number of each class in the dataset example.

Input class	Image samples	Number
Very Unclean		163
Slightly Unclean		138
Medium Clean		176
Slightly Clean		138
Very Clean		154

experiments is Windows 11, the code compilation environment is PyCharm, the Python interpreter version is 3.12.4, the

CPU is Intel(R) Xeon(R) Gold 6430, and the GPU is NVIDIA RTX A6000.

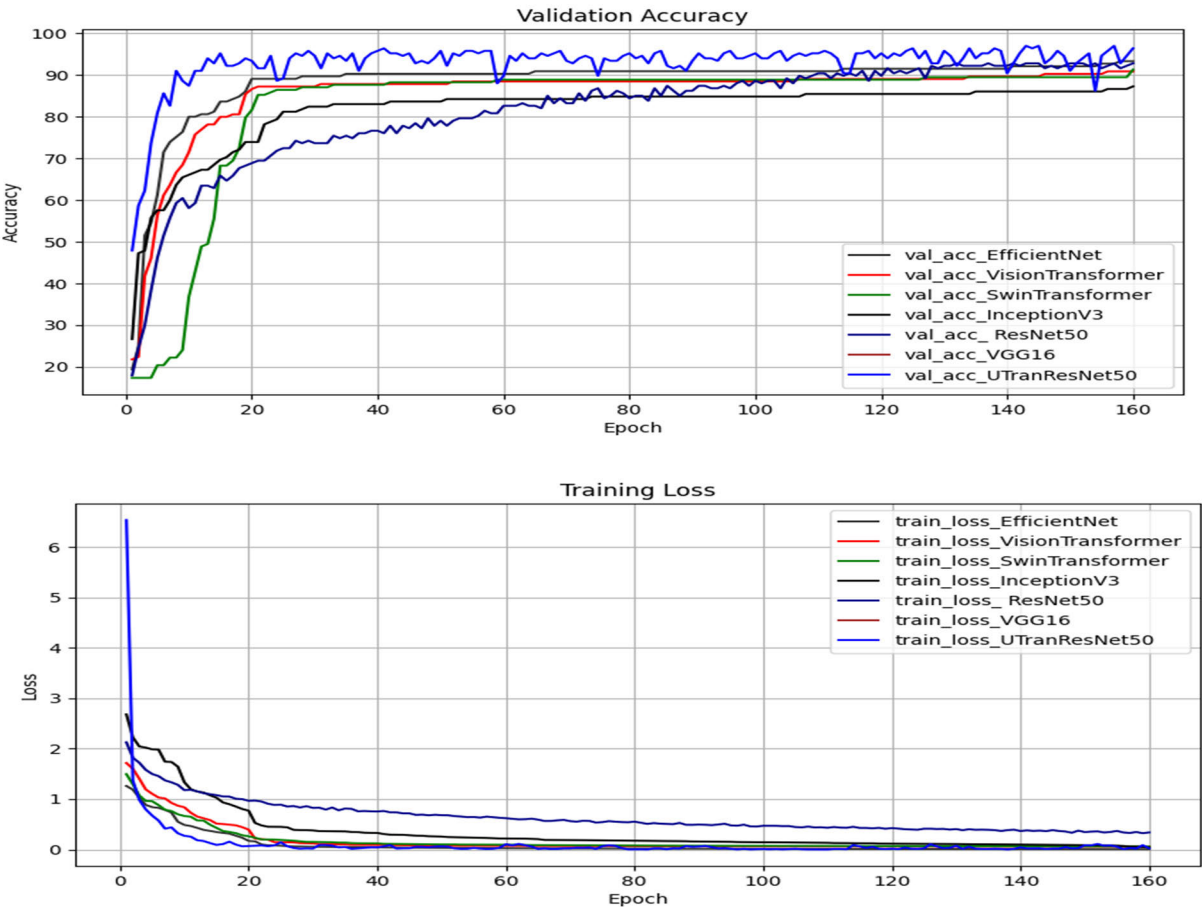


FIGURE 9. The accuracy and loss curves of models.

TABLE 3. Classification results for different network models.

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Efficient Net	93.33	93.87	92.72	93.29
Vision Transformer	90.88	91.34	90.27	90.80
Swim Transformer	91.30	91.85	91.30	91.58
Inception V3	87.27	87.80	87.27	87.53
ResNet50	92.81	92.39	93.24	92.66
VGG16	93.21	94.21	93.02	93.49
UTran-ResNet50	96.41	96.68	96.78	96.71

D. EVALUATION METRICS

In the model training, the test set data corresponding to each subset is input into the model and the model performance is evaluated using the evaluation metrics. The performance evaluation metrics of the model in this paper include Accuracy, Precision, Recall and F1 score [43]. Accuracy is the most intuitive evaluation metric, which indicates the proportion of samples correctly predicted by the model to the total samples, and we compute it as described in equation (9) to evaluate the overall performance of the classifier. Precision is a measure of the proportion of samples predicted to be positive by the

model that are positive. It indicates the ability of the classifier to correctly identify positive samples; the higher the precision, the fewer the number of false positives, and is calculated using the formula in equation (10). Recall is the proportion of all actual positive samples that are correctly predicted by the model. Calculated using the formula in equation (11), it represents the proportion of all actual positive samples that are predicted by the model, and the higher the recall, the lower the number of missed tests. F1 Score is the reconciled average of precision and recall, which integrates the balance of precision and recall, and is a commonly used evaluation

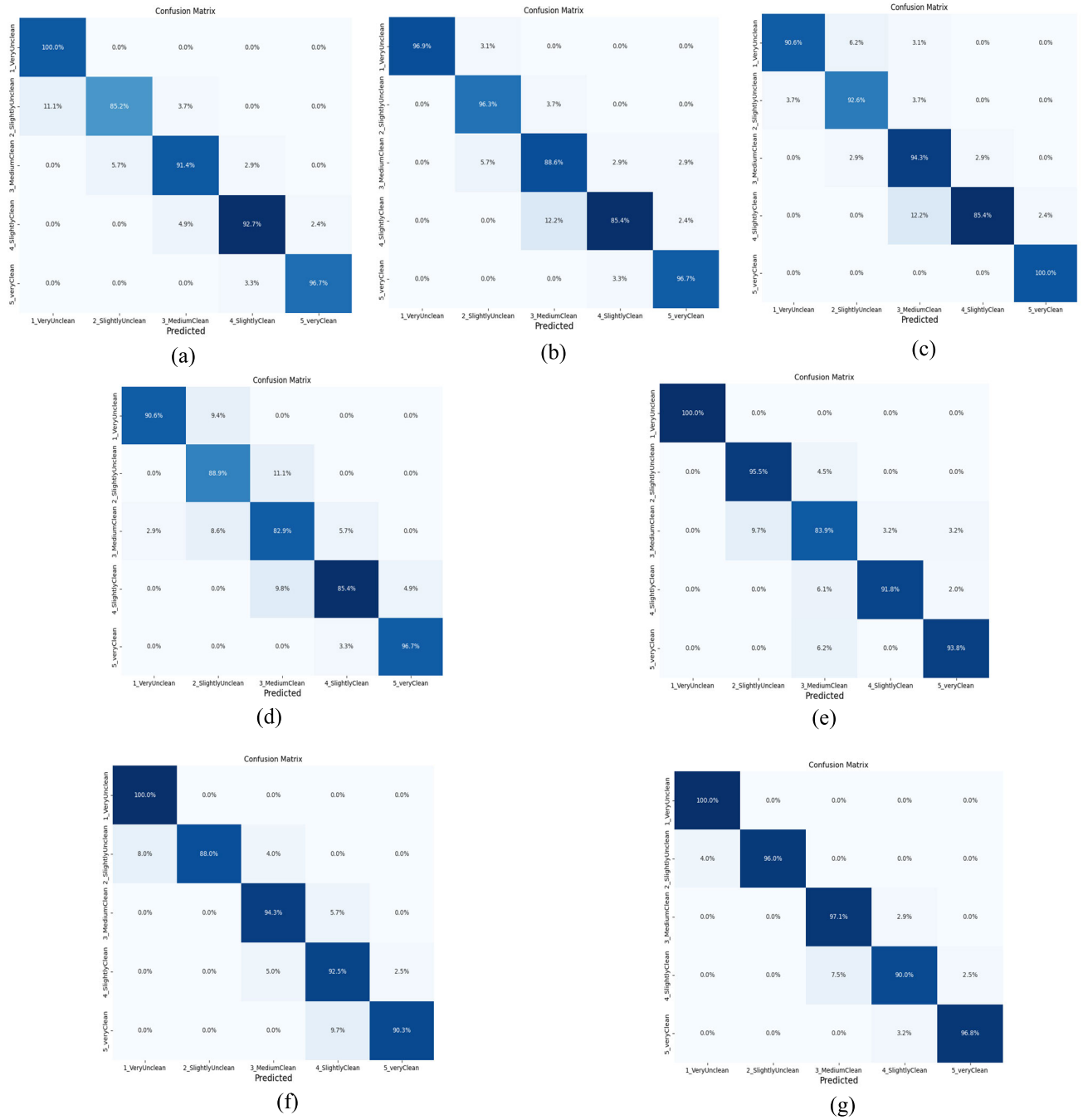


FIGURE 10. Confusion matrix for different models (a) Efficient Net, (b) Vision Transformer, (c) Swim Transformer, (d) Inception V3, (e) ResNet50, (f) VGG16, (g) UTran-ResNet50.

metric. Using the formula in equation (12), the larger the percentage is the better the model classification effect.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (10)$$

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (11)$$

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (12)$$

E. COMPARISON WITH OTHER METHODS

To evaluate the performance of the proposed UTran-ResNet50 model in recognizing dust accumulation states, this study involved training seven distinct models, including UTran-ResNet50. All networks were initialized randomly using the same dataset and hyperparameter configurations.

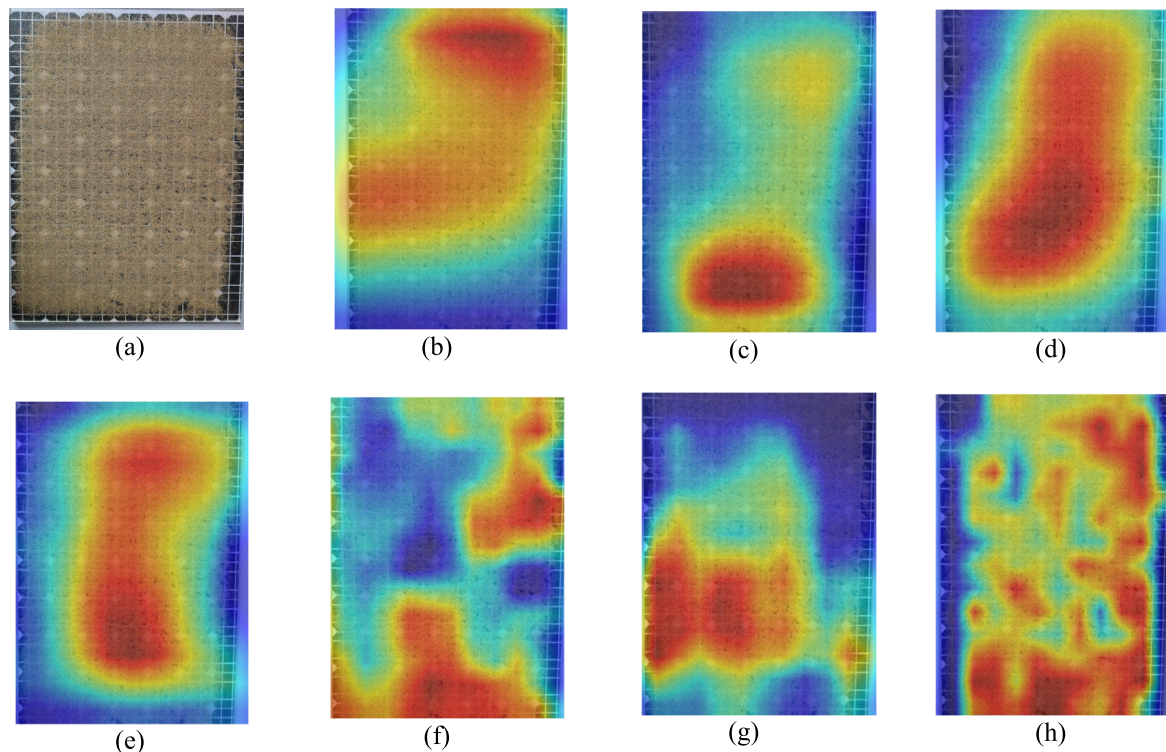


FIGURE 11. Heat maps of models:(a) original figure, (b) Efficient Net, (c) Vision Transformer, (d) Swin Transformer, (e) Inception V3, (f) ResNet50, (g) VGG16, (h) UTran-ResNet50.

Fig. 9 depicts the evolution of training loss and validation accuracy for these models across various iteration cycles.

In the training loss graph, UTran-ResNet50 exhibits the quickest decline, nearing zero, signifying superior training set fitting. Models like Efficient Net, Swin Transformer, InceptionV3, and ResNet50 achieve final losses between 0.1 and 0.5, indicating optimal fitting with stable loss curves. Vision Transformer and VGG16, however, show slower convergence and higher loss fluctuations, suggesting weaker training performance. Regarding validation accuracy, UTran-ResNet50 and ResNet50 converge rapidly with stability, while Vision Transformer and VGG16 have slower convergence and significant accuracy fluctuations, especially in later stages. Swin Transformer, Efficient Net, and InceptionV3 also demonstrate fast convergence and stable accuracy improvements. The performance of each model in the task of dust accumulation state recognition can be comprehensively assessed through the combined analysis of training loss and validation accuracy. Low training loss coupled with high validation accuracy generally implies that the model possesses good learning and generalization capabilities. The UTran-ResNet50 model performs exceptionally well in both aspects, indicating its high effectiveness in dust accumulation state recognition. The other models—Efficient Net, Swin Transformer, InceptionV3, and ResNet50—also perform well in terms of training loss and validation accuracy. Conversely, Vision Transformer and VGG16 exhibit some deficiencies in training loss and validation accuracy and may require

further optimization of the model structure or adjustment of the training strategy to enhance their performance.

Overall, the accuracy of all seven models exceeds 85%, showcasing the strong generalization and robust capabilities. The comparative experiment produces reliable results. We evaluated model performance using multiple metrics, with detailed outcomes presented in Table 3.

Fig. 10 depicts the confusion matrices of the validation set after 160-epoch training, demonstrating the proposed model's superior classification accuracy in each category compared with other networks.

To evaluate the inference efficiency of the UTran-ResNet50 model, comparative experiments were conducted with six alternative models under uniform hardware and dataset conditions. The respective inference times are documented in Table 4.

The experimental results reveal that the compatibility between hardware architecture and model computation mode significantly impacts inference efficiency. On GPUs, Vision Transformer (ViT) achieves the fastest inference by leveraging the parallelism of the global attention mechanism. Meanwhile, UTran-ResNet50 demonstrates an inference efficiency comparable to ViT on GPUs through its hybrid design combining CNN and Transformer. In contrast, UTran-ResNet50 outperforms ViT and traditional CNN models in CPU environments due to its lightweight computational requirements. For instance, ResNet50 requires 287.6ms for inference, whereas UTran-ResNet50 exhibits shorter

TABLE 4. Classification results for different network models.

Model Name	inference time on GPU	inference time on CPU
Efficient Net	95.1ms	448.2ms
Vision Transformer	49.3ms	193.6ms
Swim Transformer	61.1ms	208.5ms
Inception V3	115.8ms	513.9ms
ResNet50	104.8ms	287.6ms
VGG16	67.9ms	234.4ms
UTran-ResNet50	56.3ms	187.2ms

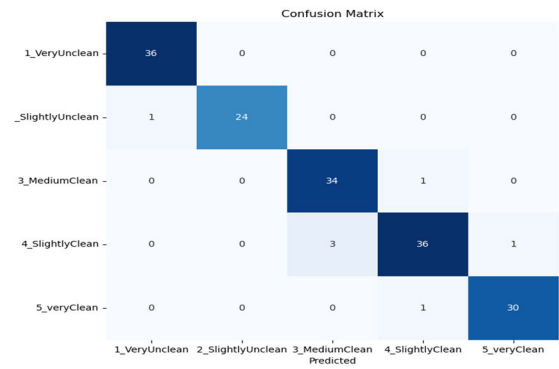
inference times. A clear trade-off exists between model complexity and real-time performance: high-complexity models, Inception V3, have the potential to improve accuracy but struggle to meet real-time demands. Conversely, UTran-ResNet50 balances local and global computation to achieve efficient cross-hardware inference under controlled complexity, offering a novel approach for edge-cloud collaborative deployment and demonstrating excellent performance and adaptability across various hardware environments.

F. GRAD-CAM-BASED VISUALIZATION RESULTS ANALYSIS

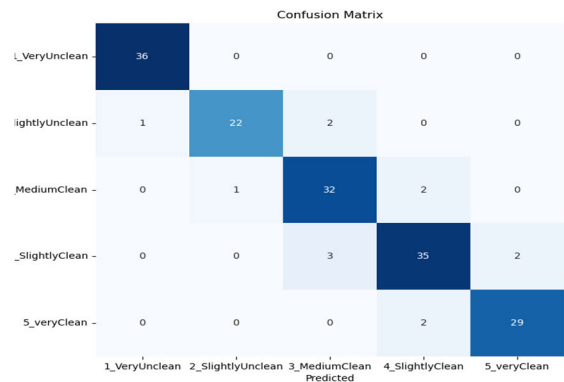
To further investigate the performance differences among various models, this study employs the Gradient-Class Activation Map (Grad-CAM) method to validate the recognition performance of the models [44], [45]. Grad-CAM generates heat maps that visualize the basis for the model's image category recognition, intuitively illustrating the decision-making foundation and its mapping relationship with category labels in the image classification task. Fig. 11 provides a comparative analysis of heat maps generated by different models. The highlighted areas indicate the key regions each model focuses on during prediction. Substantial differences exist among the models regarding their attention to dust accumulation regions. Notably, the UTran-ResNet50 model demonstrates a higher degree of focus on dust-accumulated areas compared to other models. This enhanced focus allows the model to more accurately capture essential information within the image, thereby improving its recognition accuracy and reliability.

G. ROBUSTNESS ACROSS IMAGE VARIETIES

To evaluate the model's robustness against common noise, this study conducted an adversarial robustness test in two stages. The first stage used the original images from the dataset as a baseline. The second stage utilized augmented images from the dataset and incorporated drone-captured photographs under varying flight altitudes, angles, and lighting conditions, simulating broader real-world scenarios. The sensitivity of the model to noise factors is analyzed by



(a) Use of Phase I datasets



(b) Use of Phase II datasets

FIGURE 12. Confusion matrix between the two categories.

comparing the classification accuracies of the images in these two stages. The experimental results show that the proposed model maintains high classification accuracies of 96.41% and 95.81% for both original and noise-containing images, indicating good noise robustness. Fig. 12 shows the confusion matrix between the two categories. Further analysis reveals that the model's accuracy decreases only slightly under changing light conditions, demonstrating good adaptability. However, the model's accuracy decreases significantly under background clutter, showing greater sensitivity to background interference. These findings provide critical guidance for further improvements and optimizations of the model, particularly in enhancing its robustness to complex scenes.

H. ABLATION EXPERIMENTS

Ablation study is a key part of the modeling experimentation process that can demonstrate the effectiveness of the proposed UTran-ResNet50 framework. The framework uses the ResNet50 and Transformer architecture as infrastructure. This study compares the classification results obtained with and without the use of other modules, thus providing insight into the effectiveness of the classification that will be performed by the proposed model. To validate the soundness



FIGURE 13. An analysis of ResNet50+Transformer architecture.

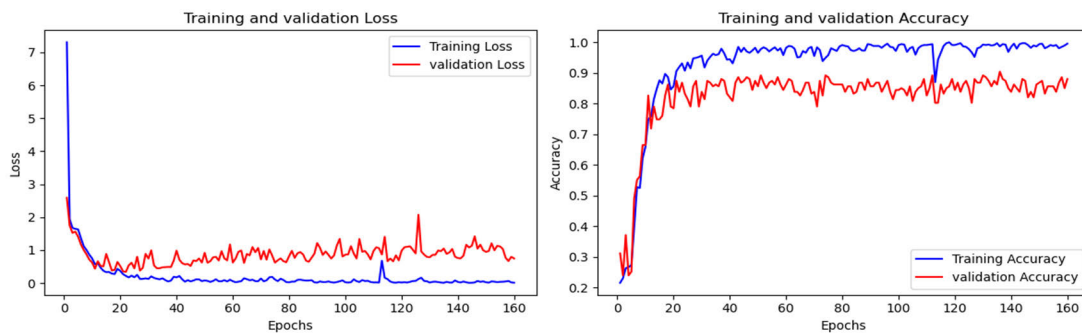


FIGURE 14. An analysis of ResNet50+Transformer+Positional Attention architecture.

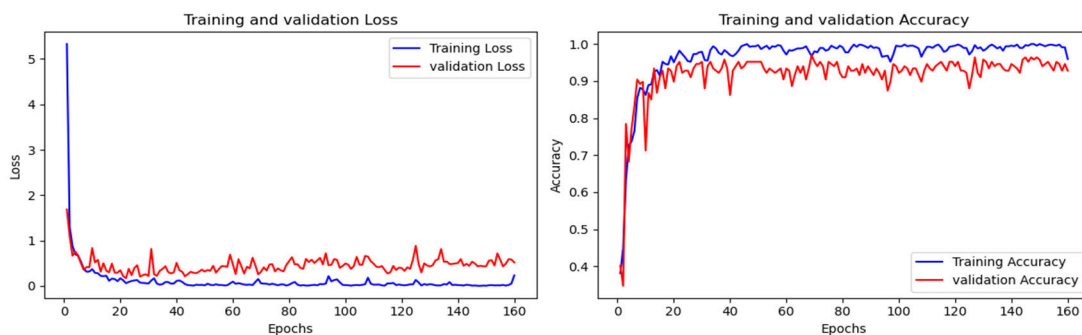


FIGURE 15. An analysis of ResNet50+Transformer+ Attention U-Net architecture.

of the modules in this paper's approach, several ablation experiments were conducted, including:

1) ANALYSIS OF ResNet50 AND TRANSFORMER

In this experimental setup, we assess the capabilities of the base model network. Fig. 13 illustrates the model's performance, achieving an accuracy of 85.03%.

2) ANALYSIS OF ResNet50, TRANSFORMER AND POSITIONAL ATTENTION

With Positional Attention and optimized parameters, the algorithm's accuracy reaches 88.02%. The Attention

mechanism alone enhances performance by capturing temporal-spatial dependencies but may suffer from reduced robustness and interpretability under image perturbations, especially in critical applications, as shown in Fig. 14.

3) ANALYSIS OF ResNet50, TRANSFORMER AND ATTENTION U-Net

The accuracy of the algorithm reaches 92.81% after using the Attention U-Net module and the parameters determined from the above experiments. The module relocates the segmented solar panels to the center of the image, which makes the model's weight distribution more balanced. However, some

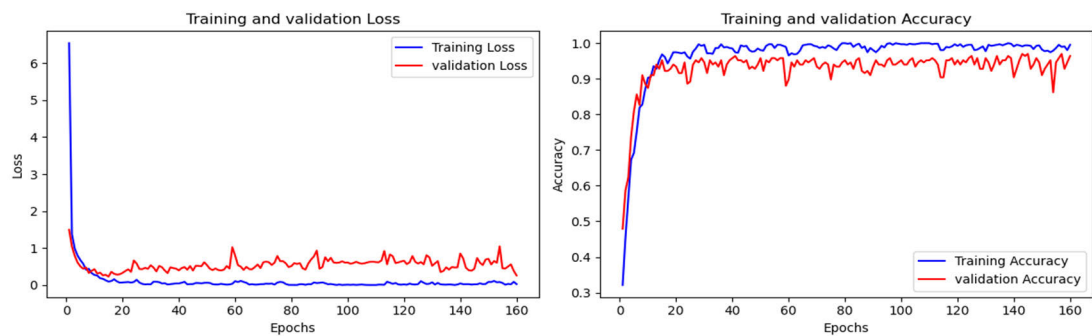


FIGURE 16. An analysis of UTran-ResNet50 architecture.

TABLE 5. Different modules results.

Model Name	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
ResNet50+Transformer	85.03	86.09	84.30	84.32
ResNet50+Transformer+Position Attention	88.02	87.86	88.03	97.90
ResNet50+Transformer+Attention U-Net	92.81	93.47	92.51	92.83

features are deleted, which may lead to the lack of some contextual information. Fig. 15 shows the performance of the model.

4) ANALYSIS OF UTran-ResNet50

By concurrently utilizing both the Attention U-Net and Positional Attention mechanisms, the model achieves an accuracy of 96.41%, representing an improvement of 11.38 percentage points over the baseline. The proposed architecture demonstrates superior performance with limited training samples, as shown on Fig. 16.

The analysis reveals that the isolated use of a single module results in only marginal improvements in network performance. A more effective strategy involves employing the attention mechanism to capture long-range dependencies after the initial pre-processing stage, where contextually rich features are extracted. This integrated approach aims to significantly enhance the overall classification capabilities of the network. This integrated approach aims to significantly enhance the overall classification capabilities of the network.

Notably, the method’s performance reaches its lowest point when no additional modules are incorporated. The proposed method’s efficacy improves with the inclusion of one module and peaks when two modules are used concurrently. This indicates that each module contributes incrementally to the overall performance enhancement. Table 5 illustrates the impact of combining backbone networks with different modular strategies on final classification performance and model complexity.

V. CONCLUSION

Energy and environmental concerns are driving the search for sustainable energy solutions, with a particular emphasis

on enhancing energy efficiency. Deserts, characterized by abundant solar resources and vast open spaces, present an ideal setting for photovoltaic (PV) installations. However, dust accumulation on PV panels poses a significant challenge to their performance. This study develops a specialized dataset of PV panel dust accumulation, simulated under realistic desert conditions, with a focus on analyzing limited sample sizes. We propose the UTran-ResNet50 architecture, which integrates ResNet50’s robust feature extraction capabilities with Transformer mechanisms to improve detail preservation and classification accuracy. The Attention U-Net and Positional Attention modules are employed to achieve precise classification. Our model demonstrates exceptional performance in identifying dust accumulation areas, achieving 96.41% accuracy on the custom dataset, significantly surpassing other models.

Moreover, the model developed in this research has proven effective in recognizing small samples of dust accumulation categories. To enhance the model’s versatility and broad applicability, it is crucial to ensure a comprehensive and diverse dataset, which will be pivotal in refining the model’s generalizability across various scenarios.

Future research will concentrate on expanding and diversifying the dataset, particularly regarding dust types, to evaluate factors influencing photovoltaic (PV) module performance. The objective is to enhance the model’s capability for dust detection under various desert conditions and to develop an automated unmanned aerial vehicle (UAV) monitoring system for efficient data collection. The study will also explore time series analysis for predictive maintenance in solar farms, aiming to improve efficiency and reduce costs. In addition, self-supervised learning techniques will be investigated for dust recognition, enabling models to adapt autonomously to

new environments. This approach is expected to enhance robustness and inform maintenance strategies for PV power plants.

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